

Theorem:

Equivalence of NFA and DFA:

* Every DFA is a NFA, consider 'L' is a language accepted by NFA, then there is a DFA that accept the same language 'L'.

* It can be implemented by using subset construction algorithm.

* In this method consider all the subsets of the states of NFA.

$$N = \{Q_N, \Sigma, \delta_N, q_0, F_N\}$$

$$D = \{Q_D, \Sigma, \delta_D, q_0, F_D\}$$

Q_D the states of DFA, Q_N all the states of NFA.

$Q_D = 2^{Q_N}$ an element of Q_D will be denoted by $[q_1, q_2, \dots, q_i]$

where $q_1, q_2, \dots, q_i$ are in Q

* Define $\delta_D([q_1, q_2, \dots, q_i], a) = p_1, p_2, \dots, p_i$

* If and only if $\delta_N(q_1, q_2, \dots, q_i, a) = p_1, p_2, \dots, p_i$

δ_D can be computed by applying δ_N to each state of δ_D represented by $[q_1, q_2, \dots, q_i]$ then taking the union we get $[p_1, p_2, \dots, p_i]$ in Q_D , then $\delta_D = (\delta_N, \times) = [q_1, q_2, \dots, q_i]$

* start state is same for both NFA and DFA.

* Every subset containing final state will be taken as final state.

* if & only if $\delta_N(q_0, x) = [q_1, q_2, \dots, q_i]$

2) Equivalence of NFA with ϵ transition into DFA:

(a)

Equivalence of ϵ -NFA to DFA:

The language L is accepted by some NFA with ϵ -transition if and only if ' L ' is accepted by some DFA.

Suppose $L = L(D)$ for some DFA convert D into ϵ -NFA transition, by adding transitions.

$\delta(q, \epsilon) = \phi$ for all states of ' q ' on input symbols. Let $E = \{Q_E, \Sigma, \delta_E, q_0, F_E\}$ be an ϵ -NFA transition.

Apply the subset construction algorithm to produce the DFA $D = \{Q_D, \Sigma, \delta_D, q_D, F_D\}$, we need to show that $L(E) = L(D)$ we show that $\delta_E(q_0, w) = \delta_D(q_0, w)$ by transition on the length of $w \Rightarrow |w| = 0$, then $w = \epsilon$

We know $\delta_E(q_0, \epsilon) = \epsilon\text{-closure}(q_0)$ we also know that $q_D = \epsilon\text{-closure}(q_0)$.

Finally for a DFA, we know that $\delta(p, \epsilon) = p$ for any state p , $\delta_D(q_0, \epsilon) = \epsilon\text{-closure}(q_0)$.

Suppose $w = xa$, where a is a final symbol, for w assume that the statement contains,

$$\delta_E(q_0, x) = \delta_D(q_0, x)$$

Let both are set of states $\{p_1, p_2, \dots, p_n\}$ we compute $\delta_E(q_0, w)$ by $\{r_1, r_2, \dots, r_m\}$, be

$$\bigcup_{i=1}^m \delta_E(p_i, a)$$

$$\text{Then } \delta_E(q_0, w) = \bigcup_{j=1}^m \epsilon\text{-closure}(r_j)$$

This $\delta_D(q_0, w) \dots$ which is $\delta_D(p_1, p_2, \dots, p_n, a)$

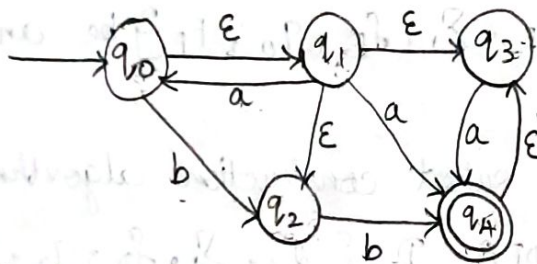
is the same set as $\delta_E(q_0, w)$, we prove that,

$$\Rightarrow \delta_E(q_0, w) = \delta_D(q_0, w)$$

Prove that a language L' is accepted by some E-NFA transition. If and only if L' is accepted by some DFA.

H.W

1) Convert E-NFA to DFA:



Step 1:

Find the ϵ -closure for all the states.

$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$

~~$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_3\}$$~~

$$\epsilon\text{-closure}(q_1) = \{q_1, q_2\}$$

~~$$\epsilon\text{-closure}(q_1) = \{q_1, q_3\}$$~~

$$\epsilon\text{-closure}(q_2) = \{q_2\}$$

$$\epsilon\text{-closure}(q_3) = \{q_3\}$$

$$\epsilon\text{-closure}(q_4) = \{q_4, q_3\}$$

Step 2:

The ϵ -closure of initial state is taken as first subset.

$$\text{First subset} = \{q_0, q_1, q_2\} \rightarrow \text{New subset ①}$$

Find the transition for first subset by using transition inputs $\{a, b\}$

$$1) \epsilon\text{-closure}(\delta(q_0, q_1, q_2), a)$$

$$\epsilon\text{-closure}(\delta(q_0, a) \cup \delta(q_1, a) \cup \delta(q_2, a))$$

$$\epsilon\text{-closure}(\emptyset \cup \{q_0, q_4\} \cup \emptyset \cup \{q_4\})$$

$$\epsilon\text{-closure}(q_0, q_4) = \{q_0, q_1, q_2, q_3, q_4\} \rightarrow \text{new subset}$$